

Determinant factorization

Basic skills

$$Q1. a) \begin{vmatrix} 6 & 2 \\ 1 & 4 \end{vmatrix} = 24 - 2 = 22$$

$$b) \begin{vmatrix} -5 & 0 \\ 1 & 2 \end{vmatrix} = -5 \times 2 = -10$$

← triangular

$$c) \begin{vmatrix} 6 & 0 & 0 \\ 4 & -5 & 0 \\ 3 & 2 & 1 \end{vmatrix} = 6 \times -5 \times 1 = -30$$

← triangular

$$Q2. |M'| = |M| = 5 \text{ as no effect on swapping}$$

$$Q3. |M'| = 2|M| = 2 \times 5 = 10$$

$$Q4. |M'| = 3^2|M| = 3^2 \times -4 = -36$$

$$Q5. |4M| = 4^3|M| = 64 \times 2 = 128$$

← 3 rows multiplied by 4

Competency skills

$$Q6. a) \begin{vmatrix} 6 & 2 \\ 1 & 4 \end{vmatrix} = k \begin{vmatrix} 12 & 4 \\ 1 & 4 \end{vmatrix} \text{ so } k = \frac{1}{2}$$

← row $\times 2$
← coefficient $\times \frac{1}{2}$ to counter $\times 2$

b) $\begin{vmatrix} 6 & 2 \\ 1 & 4 \end{vmatrix} = k \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix}$ so $k = 2$

Annotations: $\times \frac{1}{2}$ (from 6 to 3), $\times 2$ (from 1 to 1)

c) $\begin{vmatrix} 6 & 2 \\ 1 & 4 \end{vmatrix} = \begin{vmatrix} 3 & 1 \\ -3 & -12 \end{vmatrix}$ $k = -\frac{2}{3}$

Annotations: $\times \frac{1}{2}$ (from 6 to 3), $\times -3$ (from 1 to -3), so $\times 2 \times -\frac{1}{3}$

d) $\begin{vmatrix} 10 & 3 \\ 2 & -1 \end{vmatrix} = k \begin{vmatrix} 5 & 9 \\ 6 & 6 \end{vmatrix}$ $k = \frac{2}{3}$

Annotations: $\times \frac{1}{2}$ (from 10 to 5), $\times 3$ (from 2 to 6), Add rows so $\times 2 \times \frac{1}{3}$

Q7. $\begin{vmatrix} 2 & 3 & -10 \\ 4 & 10 & 6 \\ 6 & 5 & 8 \end{vmatrix} = \begin{vmatrix} 2 & 3 & -10 \\ 0 & 4 & 26 \\ 0 & -4 & 38 \end{vmatrix}$

Annotations: $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - 3R_1$

$= \begin{vmatrix} 2 & 3 & -10 \\ 0 & 4 & 26 \\ 0 & 0 & 64 \end{vmatrix}$

Annotations: $R_3 \mapsto R_3 + R_2$, $\downarrow$ triangular

$= 2 \times 4 \times 64 = 512$

Q8. $\begin{vmatrix} x & 5 & 3 \\ x+2 & -2 & 5 \\ x & 0 & 8 \end{vmatrix} = \begin{vmatrix} x & 5 & 3 \\ 2 & -7 & 2 \\ 0 & -5 & 5 \end{vmatrix}$

$$= \begin{vmatrix} x & 8 & 3 \\ 2 & -5 & 2 \\ 0 & 0 & 5 \end{vmatrix}$$

$$= \frac{1}{5} \begin{vmatrix} 5x & 8 & 3 \\ 10 & -5 & 2 \\ 0 & 0 & 5 \end{vmatrix} = \frac{1}{5} \begin{vmatrix} 5x+16 & 8 & 3 \\ 0 & -5 & 2 \\ 0 & 0 & 5 \end{vmatrix}$$

$$= \frac{1}{5} (5x+16) \times -5 \times 5 = -5(5x+16)$$

Q9.

a)

$$\begin{vmatrix} 2x-3 & 3 & 2 \\ 2x-2 & 3x+11 & 3 \\ 6x-1 & 6x-1 & 6x-1 \end{vmatrix}$$

$$= \begin{vmatrix} 2x-5 & 1 & 2 \\ 2x-5 & 3x+8 & 3 \\ 0 & 0 & 6x-1 \end{vmatrix}$$

$$= \begin{vmatrix} 2x-5 & 1 & 2 \\ 0 & 3x+7 & 1 \\ 0 & 0 & 6x-1 \end{vmatrix} = (2x-5)(3x+7)(6x-1)$$

$$b) \begin{vmatrix} x-4 & 1 & 2 \\ x+3 & x-9 & 5 \\ x-6 & x & 2x \end{vmatrix}$$

$$= \begin{vmatrix} x-3 & 1 & 2 \\ 2x-6 & x-9 & 5 \\ 2x-6 & x & 2x \end{vmatrix}$$

$$= (x-3) \begin{vmatrix} 1 & 1 & 2 \\ 2 & x-9 & 5 \\ 2 & x & 2x \end{vmatrix}$$

$$= (x-3) \begin{vmatrix} 1 & 1 & 2 \\ 0 & x-11 & 3 \\ 0 & x-2 & 2x-4 \end{vmatrix}$$

$$= (x-3)(x-2) \begin{vmatrix} 1 & 1 & 2 \\ 0 & x-11 & 3 \\ 0 & 1 & 2 \end{vmatrix}$$

$$= (x-3)(x-2) \begin{vmatrix} 1 & 2 & 2 \\ 0 & 2x-22 & 3 \\ 0 & 2 & 2 \end{vmatrix}$$

$$= (x-3)(x-2) \begin{vmatrix} 1 & 0 & 2 \\ 0 & 2x-25 & 3 \\ 0 & 0 & 2 \end{vmatrix}$$

$$= 2(x-3)(x-2)(2x-25)$$

Advanced skills

Q10. As can expand on rows/columns equivalently

$$Q11. \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$$

$$= \frac{1}{a^3} \begin{vmatrix} a^3 & a^2b & a^2c \\ a^3 & ab^2 & ac^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a^2b & a^2c \\ 1 & ab^2 & ac^2 \\ 1 & b^3 & c^3 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a^2b & a^2c \\ 0 & ab^2 - a^2b & ac^2 - a^2c \\ 0 & b^3 - a^2b & c^3 - a^2c \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a^2b & a^2c \\ 0 & ab(b-a) & ac(c-a) \\ 0 & b(b-a)(b+a) & c(c-a)(c+a) \end{vmatrix}$$

$$= a \begin{vmatrix} 1 & a^2b & a^2c \\ 0 & b(b-a) & c(c-a) \\ 0 & b(b-a)(b+a) & c(c-a)(c+a) \end{vmatrix}$$

$$= a \begin{vmatrix} 1 & a^2b & a^2c \\ 0 & b(b-a) & c(c-a) \\ 0 & 0 & c(c-a)(c+a - (b+a)) \end{vmatrix}$$

$$= a \times 1 \times b(b-a) \times c(c-a)(c-b)$$

$$= abc(b-a)(c-a)(c-b)$$

Matrix is singular when $a=0$ or $b=0$
or $c=0$ or $a=b$ or $b=c$ or $a=c$

Q12.

a) $\begin{vmatrix} x^2 & xy & x \\ y^2 & xz & y \\ z^2 & zy & z \end{vmatrix}$

$$= zx \begin{vmatrix} x & y & 1 \\ y^2 & xz & y \\ z & y & 1 \end{vmatrix}$$

$$= zx \begin{vmatrix} y^2 & xz & y \\ x & y & 1 \\ z & y & 1 \end{vmatrix} = zx \begin{vmatrix} y^2 & xz & y \\ x & y & 1 \\ z-x & 0 & 0 \end{vmatrix}$$

$$= zx (z-x \begin{vmatrix} y^2 & xz \\ xz & y \end{vmatrix})$$

$$= zx (z-x)(y^3 - x^2 z^2)$$

$$= zx (z-x)(y^{\frac{3}{2}} - xz)(y^{\frac{3}{2}} + xz)$$

$$b) \begin{vmatrix} 1 & 1 & 1 \\ 1 & \cos x & \cos^2 x \\ 1 & \sin x & \sin^2 x \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 0 & \cos x - 1 & \cos^2 x - 1 \\ 0 & \sin x - 1 & \sin^2 x - 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 0 & \cos x - 1 & -\sin^2 x \\ 0 & \sin x - 1 & -\cos^2 x \end{vmatrix}$$

$$= \begin{vmatrix} \cos x - 1 & -\sin^2 x \\ \sin x - 1 & -\cos^2 x \end{vmatrix}$$

$$= -\cos^2 x (\cos x - 1) + \sin^2 x (\sin x - 1)$$

$$= -\cos^2 x (\cos x - 1) + (\cos^2 x - 1)(\sin x - 1)$$

$$= \cos^2 x (\sin x - 1 + 1 - \cos x) - (\sin x - 1)$$

$$= \cos^2 x (\sin x - \cos x) - \sin x + 1$$